

Aditya Kumar

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SUMMARY

MS student in Computational Fluid Dynamics and Aeronautics working on GPU-accelerated DNS solvers, HPC, and reduced-order and physics-informed surrogate models for fluid and thermal systems. Builds and verifies solvers from first principles (operator identities, manufactured solutions, published benchmarks) and automates CFD, surrogate-modeling, and HPC workflows in Python, CUDA, MPI, and OpenFOAM.

EDUCATION

University of Washington

MS Aeronautics and Astronautics (Fluids Concentration), GPA: 3.88/4.0

Seattle, WA

Sep 2025 - Present

Vishwakarma Institute of Technology

BTech Mechanical Engineering, CGPA: 8.42/10 (Top 10%)

Pune, India

Jul 2021 - May 2025

TECHNICAL SKILLS

Programming: Python, MATLAB, Fortran, C, C++, CUDA, Julia, Bash/Shell Scripting, SQL

HPC and Parallel Computing: MPI, OpenMP, GPU Computing (CUDA), SLURM, PBS, Cluster Computing

CFD and Simulation: ANSYS Fluent, PyAnsys (PyFluent), STAR-CCM+, OpenFOAM, ParaView

Numerical Methods: Finite Volume Discretization, RANS/LES Turbulence Modeling, Convergence Analysis, Method of Manufactured Solutions

Surrogate Modeling: Design of Experiments, Reduced-Order Models (ROMs), Gaussian Process Regression, Physics-Informed Neural Networks (PINNs)

ML and Data Tools: PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, Matplotlib, Streamlit

CAD: SolidWorks, AutoCAD, Siemens NX

Version Control and Tooling: Git, GitHub, Poetry, Pip, Apptainer

RESEARCH EXPERIENCE

Research Assistant, Computational Fluid Mechanics Lab

University of Washington

Seattle, WA

Sep 2025 - Present

- Developing a GPU-accelerated Poisson solver (CUDA, Fortran, MPI) on an HPC cluster; benchmarking accuracy against Taylor-Green vortex decay test cases with automated Python post-processing pipelines
- Implementing RANS/LES turbulence models via finite volume discretization with convergence analysis; documenting methodology and presenting results to the research group
- Executing hybrid MPI+OpenMP distributed-memory computing for scalable numerical analysis of fluid systems

Research Co-op, Thermal Systems Engineering

Kirloskar Pneumatic Co. Ltd.

Pune, India

Jul 2024 - May 2025

- Developed Python-based 1D thermodynamic simulation models for screw compressor analysis; applied surrogate modeling and ROM techniques to improve compression efficiency by 2%
- Trained a Gaussian Process surrogate (POD dimensionality reduction, Matern kernel) on 800 to 1200 CFD cases of the compressor generated via SCORG and Simerics; ran a subset of these cases personally
- Conducted DOE studies and automated numerical workflows for thermodynamic cycle prediction; integrated data visualization pipelines with Matplotlib and Pandas, and collaborated with engineering teams on experimental validation

Aerodynamics Engineer

Veloce Racing India (Formula Student)

Pune, India

Aug 2021 - Apr 2023

- Automated multi-physics CFD workflows in ANSYS Fluent and STAR-CCM+ for external aerodynamics; achieved 5% drag reduction through computational design optimization and DOE-driven parameter studies across 30+ full-body CFD runs
- Performed pressure coefficient analysis and wake flow visualization; streamlined post-processing with ParaView scripting and Python automation
- Coordinated wind tunnel validation campaigns and mentored 5 junior engineers on CFD best practices (team finished 1st nationally, 2nd globally in the combustion category)

Manufacturing Engineering Intern

Jaguar Land Rover

Pune, India

May 2023 - Aug 2023

- Optimized Oil Filling and Cotter Press machinery through process analysis and design-for-manufacturing improvements; reduced unplanned downtime by 15%

SELECTED PROJECTS

2D Incompressible Navier-Stokes Solver

May 2026 - Present

Julia, CairoMakie (personal project)

- Built a 2D incompressible Navier-Stokes solver from scratch on a staggered MAC grid with RK3 fractional-step time integration (Le and Moin, JCP 1991); full test suite passes 15 of 15
- Validated second-order spatial accuracy via a method of manufactured solutions convergence study (L2 error slopes of 1.994, 1.999, and 2.000 across four grids) and matched centerline velocity against Ghia et al. (1982) for lid-driven cavity flow at Re 100 and Re 400
- Implemented dual pressure-Poisson solver paths (FFT with modified wavenumbers for periodic domains, conjugate gradient for wall-bounded domains) with a discrete gradient built as the negative divergence transpose for exact divergence-free projection

Conjugate Heat Transfer Study of a Liquid-Cooled Cold Plate

Apr 2026 - Aug 2026

OpenFOAM, Python, scikit-learn, SLURM, Apptainer (personal project)

- Automated a 96-case conjugate heat transfer design sweep of a liquid-cooled microchannel cold plate under a 700W GPU-class heat load via a templated SLURM array pipeline (89 of 96 cases converged)
- Trained Gaussian Process surrogate models and ran a constrained grid-search optimization, cutting thermal resistance 22% at matched pumping power; confirmed the surrogate's prediction against an independent CFD solve to five significant figures
- Validated the solver against the analytical Shah-London friction factor correlation (0.37% error) and ran a formal Grid Convergence Index study per Roache methodology (2.21% on pressure drop, 0.06% on thermal resistance)

AI-Powered Airfoil Optimization Agent

Jan 2025 - Present

Python, PyAnsys, PINNs, Streamlit (personal project, in progress)

- Implementing Physics-Informed Neural Networks to predict aerodynamic coefficients (CL, CD) from CFD databases; accelerating design evaluation 3 to 4x over ANSYS Fluent direct solve through surrogate modeling
- Automated the end-to-end workflow (mesh generation, PyAnsys/PyFluent execution, post-processing, ML training) and engineered a Streamlit app for multi-parameter airfoil optimization

Hydrodynamic Simulation of Marine Systems

Jun 2023 - Dec 2023

STAR-CCM+, Multiphase CFD, FSI (course project)

- Simulated submarine wake dynamics and buoy heave motion using overset mesh, the Volume of Fluid method, and DFBI fluid-structure interaction coupling in STAR-CCM+
- Analyzed drag and pressure distribution in the multi-phase marine environment and generated flow-field visualizations for technical review

PUBLICATIONS AND HONORS

3 manuscripts submitted to peer-reviewed international journals on CFD and thermal systems (under review)

Formula Student Program Mentor: taught ANSYS Fluent, STAR-CCM+, and CAD to 10+ students; organized CFD workshops for 50+ participants

MathWorks Modeling Competition Award

Certifications: Mathematical Modeling (IIT Kanpur), NPTEL MATLAB Fundamentals, Oracle Cloud Infrastructure Generative AI Specialization